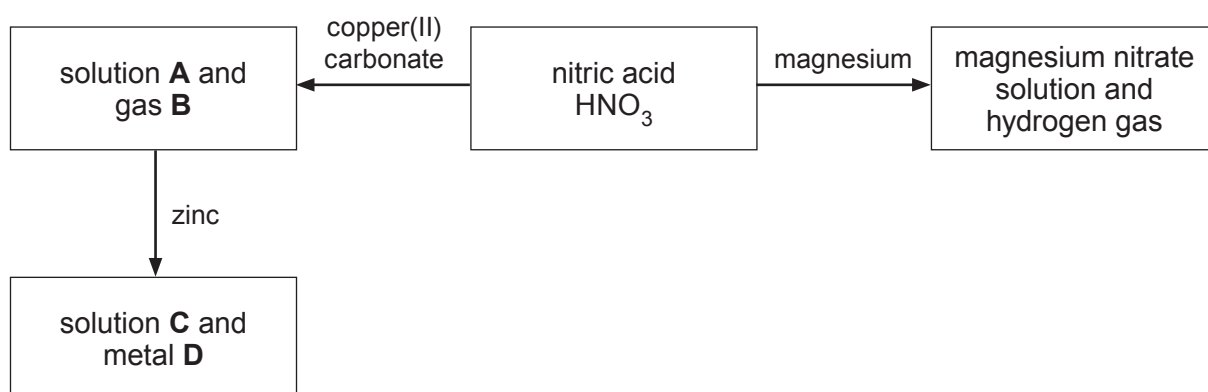


5. (a) The diagram shows a sequence of reactions from nitric acid.



- (i) Use the information in the diagram to name the following substances. [3]

solution **A**

gas **B**

solution **C**

metal **D**

- (ii) Write the balanced symbol equation for the reaction between magnesium and nitric acid. [3]

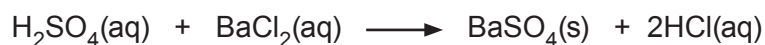
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- (iii) Name the type of reaction that takes place between solution **A** and zinc. [1]

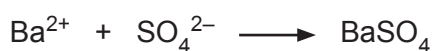
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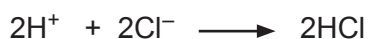
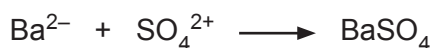


- (b) (i) The sulfate ions in sulfuric acid are identified using barium chloride solution. The equation for the reaction is shown below.



Tick (✓) the correct ionic equation for the formation of the precipitate. [1]


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- (ii) A student was asked to identify the ions present in barium chloride solution.

- I. Other than adding sulfate ions, state how the student would test for barium ions. Give the expected observation for a positive test. [1]

.....

.....

- II. State how the student would test for chloride ions. Give the expected observation for a positive test. [1]

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